Supplement for VERA: Support-Aware Shift Envelopes for Representation Erasure

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Abstract

This supplement provides full proofs, protocol details, implementation audits, and receipt-level reporting for VERA. It follows the claims and terminology of the main paper. Standard finite-family testing, exact binomial intervals, CVaR duality, and bounded-density-ratio robustness are credited as prior machinery; the proposed object is the support-aware erasure shift envelope for paired representation interventions and a registered retrained-attacker portfolio.

1 Scope and Reproducibility Contract

The exploratory parent preregistration is identified by SHA-256 `4c9343207f940220d762e230023531af353cc0886a982f133ff7794c87f13ab7`. It governs seeds 0–4, which were used only for estimand and power diagnostics. The untouched-seed balanced confirmatory preregistration is identified by SHA-256 `2767b64f1fa844f512026c5cc4d5e81bca4ed92bef9c8dfd5287dec2918395aa` and immutable Git tag `vera-balanced-confirmatory-locked-2026-07-14`. The untouched-seed secondary-ablation protocol is identified by SHA-256 `389b32fe926bf427abc80b4b249c6e9f27342b1ade3ecfc19e71fee32b1362a6`. The descriptive real learning-curve diagnostic is separately fixed by SHA-256 `b2f0a5c66c5fa55c84aeeaa30f3978204a4217d8c50e60973a671a21d316e824`. The candidate-family/group-count coverage extension is fixed by SHA-256 `c63a4a5b68ab194dd52bbaea1f1d3266a450147349e9994a51cd0566fdf61b2e`. The confirmatory lock fixes datasets, methods, candidates, seeds 5–12, splits, thresholds, attackers, validation fractions, the primary IUT at $\Gamma = 1$, the $\Gamma = 1.01$ sensitivity, the simultaneous envelope, and every pass condition before the first new-seed run.

All claim-grade real runs are tied to one parent repository commit and one pinned upstream eraser commit. Each compact JSON receipt names its split indices, preprocessing policy, official upstream entry point, candidate strengths, and the SHA-256 digest of every external-drive audit archive. The audit opens every archive and verifies required arrays, lengths, supports, attacker names, candidate uniqueness, repository cleanliness, and commit identity. A missing, proxy, dirty, or hash-mismatched row fails closed.

2 Shift-Robust Paired Certification

Let P denote the validation distribution and let $V : \mathcal{X} \to [a, b]$ be a bounded audit variable. In the target-preservation application, $V = H_e$ is the paired increase in loss caused by edit e relative to the identity edit on the same example. For $\Gamma \geq 1$, define the ambiguity set

$$\mathcal{Q}_\Gamma(P) = \left\{ Q \ll P : 0 \leq \frac{\mathrm{d}Q}{\mathrm{d}P} \leq \Gamma \quad P\text{-almost surely} \right\} \tag{1}$$

and the robust risk $\rho_{\Gamma,P}(V) = \sup_{Q \in \mathcal{Q}_\Gamma(P)} \mathbb{E}_Q[V]$. The reference law is contract-specific. Target contracts use the environment-conditional law P_g. Leakage uses P_s, the law conditional on represented source class $s \in \{0, 1\}$, and evaluates equal-weight robust class recall rather than ordinary accuracy. Accordingly, an admissible deployment profile requires $Q_g \in \mathcal{Q}_{\gamma_g}(P_g)$ for target harm and $Q_s \in \mathcal{Q}_{\eta_s}(P_s)$ for leakage. Environment mixture weights may vary arbitrarily. Source-class prevalences may also vary arbitrarily because the leakage estimand fixes the two class weights at one half; this does not imply control of prevalence-weighted ordinary accuracy.

Lemma 1 (Reweighting–CVaR identity)**.** *For every bounded V and $\Gamma \geq 1$,*

$$\rho_{\Gamma,P}(V) = \inf_{\eta \in [a,b]} \left\{ \eta + \Gamma\, \mathbb{E}_P[(V - \eta)_+] \right\}. \tag{2}$$

Thus $\rho_{\Gamma,P}$ is upper-tail conditional value at risk at level $1 - 1/\Gamma$.

Proof. Write $w = \mathrm{d}Q/\mathrm{d}P$. The primal problem maximizes $\mathbb{E}_P[wV]$ subject to $0 \leq w \leq \Gamma$ and $\mathbb{E}_P[w] = 1$. Introducing a multiplier η for the equality constraint and maximizing pointwise over w gives $\eta + \Gamma\mathbb{E}_P[(V - \eta)_+]$. Strong duality follows because $w \equiv 1$ is feasible and the bounded linear program has a finite optimum. The optimizer places the largest permitted mass on the upper tail of V. □

Given independent validation observations $V_1, \ldots, V_n$, let

$\widehat{P}_n$ be their empirical distribution and let

$$\widehat{\rho}_{\Gamma,n}(V) = \inf_{\eta\in[a,b]} \left\{ \eta + \frac{\Gamma}{n} \sum_{i=1}^{n} (V_i - \eta)_+ \right\}. \tag{3}$$

Theorem 1 (Uniform robust paired-edit certificate)**.** *Fix K bounded audit variables before observing their certification folds. Variable k has range $[a_k, b_k]$ and n_k independent observations. Define*

$$\epsilon_k = \sqrt{\frac{\log(2K/\delta)}{2n_k}}, \\ U_k(\Gamma) = \min\left\{b_k, \widehat{\rho}_{\Gamma,n_k}(V_k) + \Gamma(b_k - a_k)\epsilon_k\right\}. \tag{4}$$

Then, with probability at least $1-\delta$, simultaneously for all k and all $\Gamma \geq 1$, and all $Q \in \mathcal{Q}_\Gamma(P_k)$, $\mathbb{E}_Q[V_k] \leq U_k(\Gamma)$. Consequently, any data-dependent edit selected only when its paired target-harm bound is at most τ and all of its preregistered attacker-leakage bounds are at most λ satisfies those contracts for every allowed deployment distribution.

Proof. For one variable, the Dvoretzky–Kiefer–Wolfowitz inequality gives $\sup_t |F_n(t) - F(t)| \leq \epsilon_k$ except on an event of probability δ/K. For every $\eta \in [a_k, b_k]$,

$$\begin{aligned} &\left|\mathbb{E}_P[(V-\eta)_+] - \mathbb{E}_{\widehat{P}_n}[(V-\eta)_+]\right| \\ &\quad = \left|\int_\eta^{b_k} (F_n(t) - F(t))\,\mathrm{d}t\right| \\ &\quad \leq (b_k - a_k)\epsilon_k. \end{aligned} \tag{5}$$

Applying this inequality inside Lemma 1 and taking the infimum over η yields $\rho_{\Gamma,P}(V) \leq \widehat{\rho}_{\Gamma,n}(V) + \Gamma(b_k - a_k)\epsilon_k$. A union bound over K audit variables proves simultaneous validity. On this event, every accepted edit obeys each contract; selection among accepted edits does not change the event. □

Corollary 1 (Exact discrete paired certificate)**.** *Let $L \in \{0,1\}$ have success probability p, and let $H \in \{-1,0,1\}$ have probabilities (p_-, p_0, p_+). Then*

$$\rho_{\Gamma,P}(L) = \min\{1, \Gamma p\}, \tag{6}$$

and, writing $a = \Gamma p_+$ and $d = \Gamma(1-p_-)$,

$$\rho_{\Gamma,P}(H) = \begin{cases} 1, & a \geq 1, \\ a, & a < 1 \leq d, \\ a + d - 1, & d < 1. \end{cases} \tag{7}$$

For a registered family of K contracts, replace p by a one-sided Clopper–Pearson upper bound at error δ/K. For paired harm, let U_+ be a one-sided upper bound for p_+ and L_- a one-sided lower bound for p_-, each at error $\delta/(2K)$, and substitute $\widetilde{p}_+ = \min\{U_+, 1-L_-\}$ and $\widetilde{p}_- = L_-$ in the formula. The clipping preserves the probability-simplex constraint and can only tighten the upper bound. These substitutions give simultaneous upper confidence curves for every $\Gamma \geq 1$. Inverting the curves therefore gives the same shift-radius and false-acceptance guarantees as Theorem 3.

Proof. The worst density ratio assigns weight Γ first to value one, then to value zero, and finally assigns the residual unit mass to value minus one. This greedy allocation yields the displayed formulas. The Bernoulli expression is increasing in p. The paired expression is increasing in p_+ and decreasing in p_- in every branch. On the joint confidence event, $p_+ \leq U_+$ and $p_- \geq L_-$. Since also $p_+ \leq 1-p_- \leq 1-L_-$, $p_+ \leq \widetilde{p}_+$; hence the clipped substitution dominates the true robust risk. A union bound gives simultaneous validity. The coverage event is independent of Γ, so it holds over the entire continuum and remains valid after inversion and edit selection. □

For attacker a, let $L_{e,a} = \mathbf{1}\{a_e(Z^e) = S\}$ and define the balanced robust leakage

$$B_{e,a}(\eta_0, \eta_1) = \frac{1}{2} \sum_{s\in\{0,1\}} \rho_{\eta_s, P_s}(L_{e,a}). \tag{8}$$

If $U_{e,a,s}(\eta_s)$ is an upper confidence curve for the class-s recall, then $\widehat{B}_{e,a} = \frac{1}{2}(U_{e,a,0} + U_{e,a,1})$ upper-bounds Eq. (8) on the intersection of the two coverage events. A component-level error budget α is therefore obtained by using one-sided Clopper–Pearson bounds at $\alpha/2$ for the two recalls.

Lemma 2 (Source-prior invariance)**.** *Let the deployment model declare both source-conditional laws and let source prevalence be any $\pi \in [0,1]$, including 0 or 1. Suppose the two source-conditional deployment laws obey $Q_s \in \mathcal{Q}_{\eta_s}(P_s)$. The balanced robust leakage in Eq. (8) is independent of π and is at most λ whenever $\widehat{B}_{e,a} \leq \lambda$. A constant binary attacker has balanced leakage $1/2$ at $\eta_0 = \eta_1 = 1$, regardless of which class it always predicts.*

Proof. The estimand assigns weight one half to each conditional recall by definition, so prevalence π does not enter it. On the joint confidence event, each robust recall is at most its corresponding upper curve; averaging gives the first claim. A constant attacker has class recalls $(1,0)$ or $(0,1)$ under IID evaluation, whose equal-weight mean is $1/2$. At $\pi \in \{0,1\}$, the unused conditional remains part of the declared standardization model; the balanced functional is not ordinary accuracy observable from a deployment sample containing only one class. □

Theorem 2 (Fixed-profile candidate intersection–union test)**.** *Fix M candidate edits before certification. For candidate e, let $\{R_{e,j} \leq c_j\}_{j\in\mathcal{J}_e}$ contain every target and balanced leakage component at one declared shift profile. Suppose each reported component bound $U_{e,j}$ undercovers $R_{e,j}$ with probability at most $\alpha = \delta/M$. For paired target harm, the positive- and negative-count bounds split this α equally; for balanced leakage, its two class-recall bounds do the same. Accept candidate e only if $U_{e,j} \leq c_j$ for every j, then select any accepted candidate by a fixed utility rule. The probability of deploying any candidate that violates at least one component contract is at most δ. If identity is itself a selectable deployment action, it is included among the M candidates; using identity only to define paired harm does not add a selection hypothesis.*

Proof. For each unsafe candidate e, choose one population-violated component $j(e)$ before the certification draw. False acceptance of e implies $U_{e,j(e)} \leq c_{j(e)} < R_{e,j(e)}$, an undercoverage event of probability at most α. No union bound across components is needed because acceptance requires every component to pass: this is an intersection–union test. A union bound over the at most M unsafe candidates gives $M\alpha = \delta$. Selection among accepted candidates is contained in the same event. □

Corollary 2 (Worst-group mixture shift)**.** *Let P_g be validated group-conditional distributions for $g \in \mathcal{G}$. If $\rho_{\Gamma,P_g}(V_e) \leq c$ is simultaneously certified for every edit e and group g, then for every collection $Q_g \in \mathcal{Q}_\Gamma(P_g)$ and every deployment mixture $Q_\pi = \sum_g \pi_g Q_g$ with π in the probability simplex, $\mathbb{E}_{Q_\pi}[V_e] \leq c$.*

Proof. $\mathbb{E}_{Q_\pi}[V_e] = \sum_g \pi_g \mathbb{E}_{Q_g}[V_e] \leq \sum_g \pi_g c = c$. □

Let $U^T_{e,g}(\gamma_g)$ be a simultaneous upper curve for paired target harm in observed environment g, and let $U^L_{e,a,s}(\eta_s)$ be one for attacker a's class-s recall. The *support-aware erasure shift envelope* is

$$\widehat{\mathcal{E}}_e = \{(\boldsymbol{\gamma}, \boldsymbol{\eta}) : U^T_{e,g}(\gamma_g) \leq \tau \quad \forall g, \\ \tfrac{1}{2} \textstyle\sum_s U^L_{e,a,s}(\eta_s) \leq \lambda \quad \forall a\}. \tag{9}$$

Profiles that place deployment mass on an unobserved required environment are excluded. A missing source class excludes balanced-leakage certification unless a structural assumption identifies its conditional recall.

Corollary 3 (Support-aware erasure shift envelope)**.** *With probability at least $1-\delta$, simultaneously for every registered edit e, every deployment profile in its certified envelope satisfies every registered target and balanced-leakage contract. Environment mixture weights and source-class prevalences may be chosen arbitrarily after certification. The largest certified common budget is*

$$\widehat{\Gamma}_e = \sup\{\Gamma \geq 1 : (\Gamma \mathbf{1}_{|\mathcal{G}|}, \Gamma, \Gamma) \in \widehat{\mathcal{E}}_e\}, \tag{10}$$

with value zero if the IID profile fails. No additional multiplicity correction is required to inspect the envelope or choose a contained profile after certification.

Proof. The simultaneous event used for the individual curves already ranges over every edit, target environment, source class, and attacker. On that event, membership in Eq. (9) controls target harm for every allowed Q_g and balanced leakage for every allowed Q_s. Arbitrary environment mixing preserves the target threshold by Corollary 2; source prevalence does not enter Eq. (8). Restricting to the diagonal profile gives the common radius. Inspecting or intersecting objects that are all valid on the same event does not spend additional error probability. □

Corollary 4 (False acceptance)**.** *Under the conditions of Theorem 1, the probability that VERA accepts an edit violating at least one declared robust contract is at most δ.*

For edit e, let $\mathcal{K}_e$ index its paired target-harm and registered leakage contracts, with thresholds c_k. Define the population erasure shift radius

$$\Gamma^\star_e = \sup\{\Gamma \geq 1 : \rho_{\Gamma,P_k}(V_k) \leq c_k \text{ for every } k \in \mathcal{K}_e\}, \tag{11}$$

with $\Gamma^\star_e = 0$ when a contract fails at $\Gamma = 1$. Let $\widehat{\Gamma}_e$ replace each population robust risk by the upper bound in Theorem 1, and use the same zero convention.

Theorem 3 (Certified erasure shift radius)**.** *With probability at least $1-\delta$, simultaneously for every registered edit e, $\widehat{\Gamma}_e \leq \Gamma^\star_e$. The statement holds uniformly over the continuum of $\Gamma \geq 1$; consequently, a user may select any deployment budget $\Gamma \leq \widehat{\Gamma}_e$ after observing the certificate without an additional multiplicity correction. If computation is capped at $\Gamma_{\max}$, equality $\widehat{\Gamma}_e = \Gamma_{\max}$ is reported as right-censored.*

Proof. The DKW event in Theorem 1 does not depend on Γ. On that event, for every k and every $\Gamma \geq 1$, the reported upper curve dominates $\rho_{\Gamma,P_k}(V_k)$. Hence every Γ admitted by all empirical upper curves is admitted by all population contracts. Taking the supremum proves $\widehat{\Gamma}_e \leq \Gamma^\star_e$ for every edit on the same simultaneous event. Robust risk is nondecreasing in Γ because the ambiguity sets are nested, so the feasible budgets form an interval and bisection is valid. Post-certificate choice within that interval leaves the event unchanged. □

Theorem 4 (Unsupported-cell impossibility)**.** *Fix an edit e, a contract threshold c, and let O_n contain every random quantity available to a validation-only protocol. Suppose an audit cell A has zero validation probability, the allowed deployment class contains a law Q_A supported on A, and the nonparametric model class contains two worlds M_0, M_1 such that*

1. *O_n has the same distribution under M_0 and M_1;*
2. *$\mathbb{E}_{Q_A,M_0}[V_e] \leq c$; and*
3. *$\mathbb{E}_{Q_A,M_1}[V_e] > c$.*

If a possibly randomized protocol controls false acceptance at level δ uniformly over this model and deployment class, then its probability of accepting e under the safe world M_0 is at most δ.

Proof. Include the protocol's independent random seed in O_n. By condition 1, its acceptance probability p is identical in the two worlds. Under M_1 and Q_A, acceptance of e is a false acceptance by condition 3. Uniform control therefore requires $p \leq \delta$. The same bound holds under M_0, although condition 2 says that e is safe there. Equivalently, the two observable experiments have total-variation distance zero, so no test can distinguish them. □

An audit cell may be an input region, an environment, or an environment–source-label pair. Thus the result covers both an absent deployment environment and a conditional leakage contract whose required label cell is unobserved. A single-class slice cannot identify the omitted class's conditional leakage without a structural assumption. In the preregistered Camelyon17 split, official hospital center 2 occurs only in

M	$\lvert\mathcal{G}\rvert$	False accepts	Max. CP/δ
5	1	0	0.418
5	3	0	0.418
5	5	0	0.418
9	1	0	0.418
9	3	0	0.418
9	5	0	0.418
13	1	17	0.418
13	3	0	0.418
13	5	0	0.418
17	1	11	0.541
17	3	0	0.418
17	5	0	0.418

Table 1: Independent coverage stress test over candidate-family and validated-group counts. The final column is the largest simultaneous one-sided 95% Clopper–Pearson upper bound divided by its registered δ; values at most one pass.

the external split, whereas centers 0, 1, 3, and 4 are observed during certification; moreover, the binary center-0 source label is constant in the center-2 external slice. A uniform certificate over the missing cells must acquire representative support, assert a testable cross-cell model, or abstain. The dataset instantiates the support boundary; it is not evidence for the theorem by itself.

3 Exact Finite-Sample Study

The exact study was locked before execution. It crosses certification sizes $n \in \{250, 500, 1000, 2000, 5000, 10000\}$ with $\delta \in \{0.01, 0.05, 0.10\}$ and $\Gamma \in \{1, 1.01, 1.25\}$, using 2,000 independent replicates in each of 54 cells. Twelve candidates trade paired target harm in three environments against four attackers, each measured by two source-class recalls. The fixed-profile IUT allocates $\delta/12$ to each candidate; paired harm and balanced leakage split each component allocation between their two exact one-sided bounds.

The finite multinomial/binomial law supplies the exact acceptance probability, simultaneous prediction bands, and false-acceptance bounds. All 54 cells pass: zero false acceptances occur in 108,000 repetitions, and every observed abstention rate falls inside its simultaneous exact prediction band. The largest absolute prediction error is 0.0143. This study validates the discrete implementation under a known law. It does not establish novelty or replace the real benchmark study.

4 Candidate-Family and Group-Count Coverage Grid

A separately locked extension crosses six validation sizes, four candidate-family sizes, three validated-group counts, and three δ levels. Every cell uses 2,000 Monte Carlo repetitions and contains at least one truly violating candidate. Each family counts the identity control action in addition to its eraser candidates.

5 Official Real-Study Protocol

The real study crosses INLP, R-LACE, LEACE, TaCo, and MANCE++ with Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, Bios, and GaitPDB at untouched seeds 5–12. Candidate construction uses at most 8,000 eraser-training examples and 2,000 construction examples; certification and external evaluation each use at most 8,000 examples. Standardization and randomized PCA are fitted on eraser-training data only. Every edit retrains a balanced logistic target probe and four leakage attackers: linear logistic, RBF-Nystroem logistic, random forest, and a two-layer MLP.

The registered grid uses certification fractions $\{0.05, 0.10, 0.25, 0.50, 1.00\}$, paired target-harm thresholds $\tau \in \{0.05, 0.10, 0.20\}$, leakage thresholds $\lambda \in \{0.60, 0.70, 0.80\}$, $\delta = 0.05$, primary budget $\Gamma = 1$, shifted sensitivity $\Gamma = 1.01$, and radius cap $\Gamma_{\max} = 8$. The decision rules are always deploy, point-feasible validation selection, VERA-IUT, the VERA simultaneous envelope, and an external oracle used only for evaluation.

Camelyon17 center 2 is absent from certification, while centers 0, 1, 3, and 4 are observed. Its center-0-versus-other binary source label is constant in the center-2 external slice. The primary protocol therefore assigns center 2 a zero envelope coordinate and abstains for a declared five-center deployment. This is procedural non-certifiability, not by itself a measured external contract violation.

6 Statistical Integrity

Threshold pairs and nested fractions share the same eight seed-level fits and examples. Configuration-level Clopper–Pearson and McNemar calculations are retained as preregistered diagnostics, not treated as independent-trial inference. The primary comparison averages each rule's nine outcomes within seed and applies the preregistered exact one-sided sign-flip test over eight seed blocks, with Holm correction over four externally estimable datasets. All adjusted values are reported regardless of outcome. Seeds 0–4 contribute to none of these estimates or tests.

External labels are unused by edit construction and candidate selection. The unattended runner nevertheless computes external arrays run by run after each candidate is fixed; aggregate analysis is deferred until the entire official matrix and receipt audit finish. The paper does not claim that external labels were physically inaccessible throughout execution.

7 Receipt-Level Confirmatory Results

Safe-retention uncertainty: seed-cluster bootstrap 95% interval 27.5%–34.5%; configuration-level Clopper–Pearson diagnostic 23.5%–39.0%. The maximum supported dataset–seed VERA violation rate was 0.0%.

8 Secondary Sensitivity Analyses

9 Claim Boundary

VERA does not prove perfect erasure, security against every measurable attacker, fairness for unmeasured concepts, clinical safety, or validity on new support. The density-ratio budgets are declared sensitivity parameters, not facts estimated

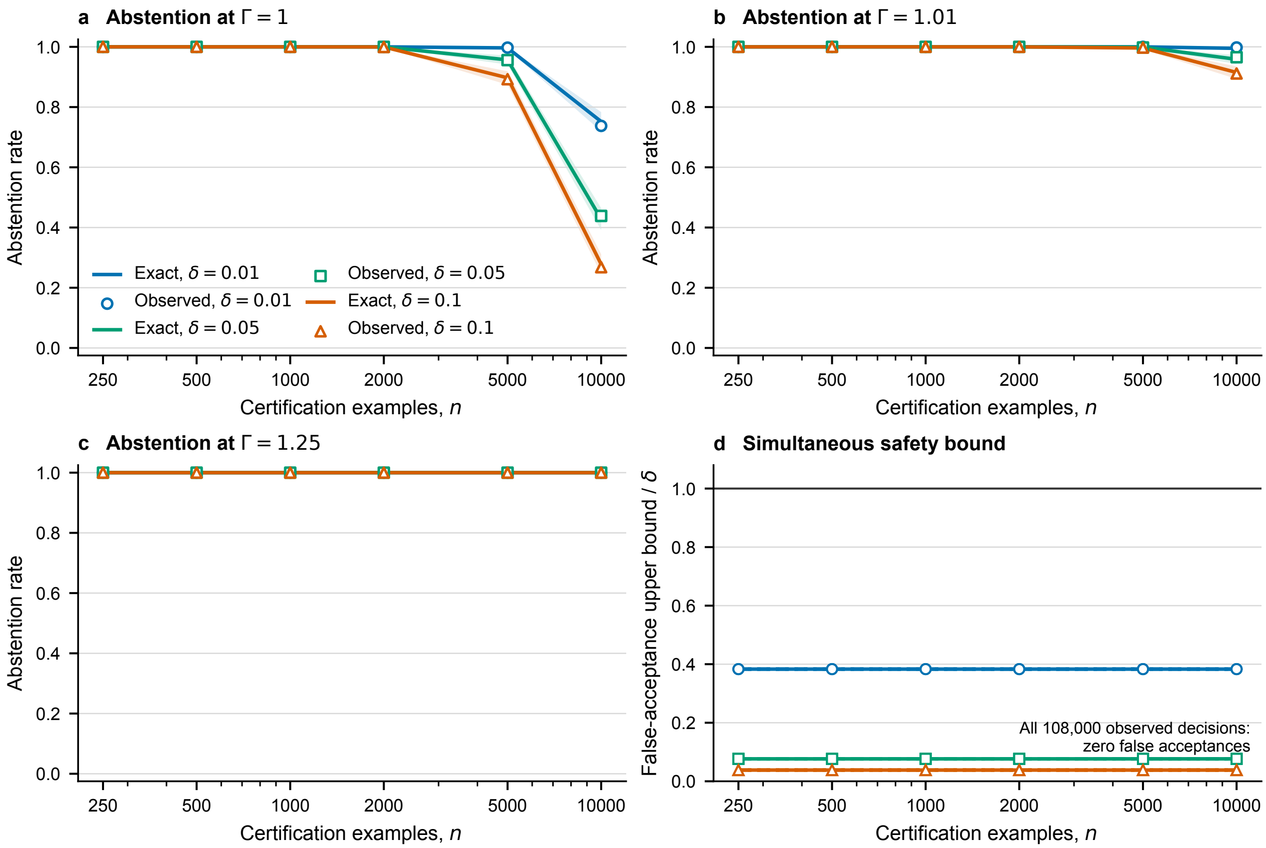

Figure 1: Exact predicted and observed abstention across the 54 locked balanced cells. Bands are simultaneous over the registered grid.

from an external test set. Internal audits check consistency and provenance but cannot establish novelty, proof correctness, or conference acceptance. Two cold reviews from researchers who publish in machine learning remain a mandatory submission gate.

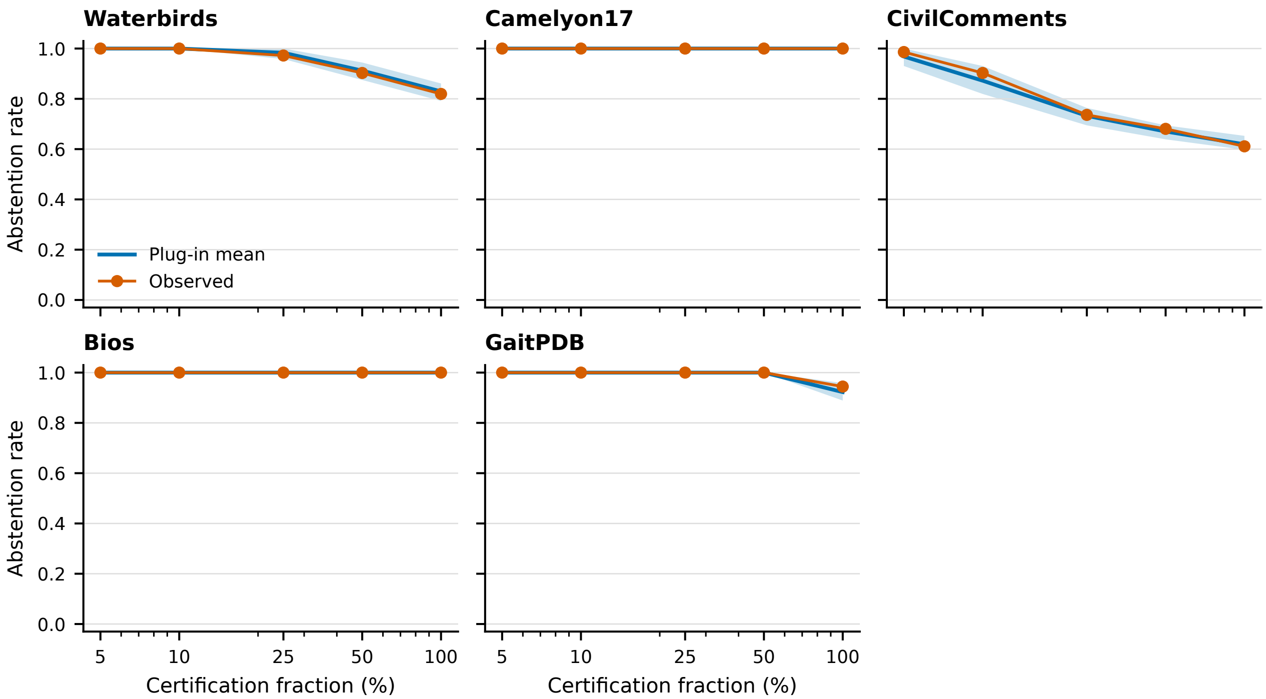

Figure 2: Observed real-data abstention versus a full-certification empirical plug-in bootstrap. Resampling preserves registered strata and aligned dependence across candidates and attackers. Unlike Fig. 1, this is a descriptive calibration diagnostic because the same full certification fold estimates its bootstrap law.

Dataset	Rule	Deploy	Viol./estimable	Cond. viol.	Unsupported
Bios	Always deploy	72/72	71/72	98.6%	0
Bios	Point selection	0/72	0/72	NA	0
Bios	VERA-IUT	0/72	0/72	NA	0
Bios	VERA envelope	0/72	0/72	NA	0
Bios	External oracle	1/72	0/72	0.0%	0
Camelyon17-WILDS	Always deploy	72/72	0/0	NA	72
Camelyon17-WILDS	Point selection	23/72	0/0	NA	23
Camelyon17-WILDS	VERA-IUT	0/72	0/0	NA	0
Camelyon17-WILDS	VERA envelope	0/72	0/0	NA	0
Camelyon17-WILDS	External oracle	0/72	0/0	NA	0
CivilComments-WILDS	Always deploy	72/72	42/72	58.3%	0
CivilComments-WILDS	Point selection	38/72	5/72	13.2%	0
CivilComments-WILDS	VERA-IUT	28/72	0/72	0.0%	0
CivilComments-WILDS	VERA envelope	27/72	0/72	0.0%	0
CivilComments-WILDS	External oracle	37/72	0/72	0.0%	0
GaitPDB	Always deploy	72/72	3/72	4.2%	0
GaitPDB	Point selection	66/72	3/72	4.5%	0
GaitPDB	VERA-IUT	4/72	0/72	0.0%	0
GaitPDB	VERA envelope	0/72	0/72	NA	0
GaitPDB	External oracle	72/72	0/72	0.0%	0
Waterbirds	Always deploy	72/72	50/72	69.4%	0
Waterbirds	Point selection	38/72	6/72	15.8%	0
Waterbirds	VERA-IUT	13/72	0/72	0.0%	0
Waterbirds	VERA envelope	9/72	0/72	0.0%	0
Waterbirds	External oracle	36/72	0/72	0.0%	0

Table 2: Full-size primary results. Each dataset has $8 \times 9 = 72$ threshold–seed configurations per rule. Unsupported counts flag deployment into the registered Camelyon17 support mismatch; they are not recoded as measured violations.

Dataset	Point-only	VERA-only	Sign-flip Holm	Sign-test Holm	McNemar Holm
Bios	0	0	1.0000	1.0000	1.0000
CivilComments-WILDS	5	0	0.5000	0.5000	0.1875
GaitPDB	3	0	1.0000	1.0000	0.5000
Waterbirds	6	0	0.5000	0.5000	0.1250

Table 3: Inference diagnostics. Sign-flip is the preregistered seed-blocked test. The sign test is a symmetry-free, lower-power secondary sensitivity. McNemar counts share fits and thresholds and are descriptive only. All values are Holm-adjusted over the four externally estimable datasets.

Official eraser	Candidate points	Run receipts
INLP	4	40
LEACE	1	40
MANCE++	1	40
RLACE	2	40
TaCo	4	40
Total	12	200

Table 4: Every candidate is generated by pinned official upstream code. One run receipt contains the complete registered strength frontier for one dataset, method, and seed; there are no proxy rows.

Dimension	Condition	Deploy	Measured violation	Safe retention
fixed profile shift budget	gamma=1	12.5%	0.0%	30.8%
fixed profile shift budget	gamma=1.01	10.3%	0.0%	25.3%
fixed profile shift budget	gamma=1.05	4.4%	0.0%	11.0%
fixed profile shift budget	gamma=1.1	0.6%	0.0%	1.4%
fixed profile shift budget	gamma=1.25	0.0%	0.0%	0.0%
fixed profile shift budget	gamma=1.5	0.0%	0.0%	0.0%
fixed profile shift budget	gamma=2	0.0%	0.0%	0.0%
fixed profile shift budget	gamma=4	0.0%	0.0%	0.0%
attacker portfolio	drop_forest	12.5%	0.0%	30.8%
attacker portfolio	drop_linear	13.1%	0.0%	32.2%
attacker portfolio	drop_mlp	25.0%	14.9%	32.2%
attacker portfolio	drop_rbf	12.8%	0.0%	31.5%
attacker portfolio	full	12.5%	0.0%	30.8%
attacker portfolio	linear_only	43.6%	34.0%	40.4%
attacker portfolio	linear_rbf	27.2%	17.7%	32.2%
eraser frontier	all five official erasers	12.5%	0.0%	30.8%
eraser frontier	leave INLP out	8.9%	0.0%	29.4%
eraser frontier	leave LEACE out	12.2%	0.0%	30.1%
eraser frontier	leave MANCE++ out	12.5%	0.0%	30.8%
eraser frontier	leave R-LACE out	12.5%	0.0%	31.2%
eraser frontier	leave TaCo out	9.2%	0.0%	25.4%
frontier granularity	coarse_5	11.4%	0.0%	31.8%
frontier granularity	full_12	12.5%	0.0%	30.8%

Table 5: Outcome-blindly locked secondary analyses. External safety always uses all four attackers. Candidate alpha and envelope multiplicity remain fixed at the original 12-candidate family, so removing attackers or erasers receives no confidence-allocation windfall.

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